

SGT HYDROEDGE

GreenDrive

Commercial Fleet Hydrogen Fuel Assist

SGT

Credibility Dossier | Customer & Investor Edition

4–9% Guaranteed ROI Threshold Cost < 4% of annual fuel spend	6–22% Real-World Fuel Savings Customer-verified results	3 Lakh+ km On-Road Validated 36 months of field testing	ARAI Published Research Paper All diesel engine types
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Coca-Cola 4-Month Enterprise POC 6–9% fuel reduction baseline	BLR Logistiks (I) Ltd 22–23% CAN-verified improvement	3 International Markets Malaysia · Uganda · Australia	1 Year ROI Guarantee Design For vehicles > 1500 L/month
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CHFA TECHNOLOGY GUARANTEE Every diesel engine operating within Zone B receives measurable fuel savings, CO ₂ reduction, emission reduction, and continuous engine decarbonisation — independent of driver behaviour.

Focused exclusively on commercial diesel vehicles consuming ≥ 1,500 litres/month
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1. The Challenge — and Why We Are Honest About It

Why logistics is different from industrial applications — and why that matters.

In a kiln, boiler, or DG set, combustion conditions are controlled, load is predictable, and the technology operates in isolation. In a commercial vehicle, a human driver sits between the technology and the outcome.

SGT has conducted more than 36 months of real-world testing across trucks, buses, and commercial vehicles. We have built customer relationships, published research, and completed enterprise POCs with organisations like Coca-Cola. We are honest about what we have learned:

- **Fuel theft** by drivers directly reduces the measurable savings the technology creates. It is a real and common challenge in the Indian logistics ecosystem.
- **Driver behaviour** — aggressive acceleration, excessive idling, improper gear usage — accounts for up to 30% variation in mileage across identical vehicles on identical routes.
- **Fuel adulteration** varies by geography and affects combustion quality independently of HHO dosing.
- **Payment discipline** and service access in a fragmented market of individual truck owners is operationally complex.

This is why investors have historically kept their distance from this category.

SGT's response is not to overclaim — it is to engineer around the problem. Our commercial strategy is built on a single, defensible principle:

The system costs less than 4% of a vehicle's annual fuel expenditure.

At that price point, every vehicle owner achieves payback within 12 months — even if the real-world saving is only 4%. The actual savings consistently measured by our customers range from 6% to 22%.

The technology guarantee: CHFA-optimised combustion delivers measurable benefit to every diesel engine in Zone B — fuel saved, CO₂ reduced, emissions cleaner, engine decarbonised. That is not dependent on driver behaviour.

2. The GreenDrive Technology — How It Works

GreenDrive is SGT HydroEdge's proprietary Controlled Hydrogen Fuel Assist (CHFA) system for diesel vehicle engines. It generates oxyhydrogen gas (HHO) on demand via alkaline water electrolysis and injects it through the vehicle's air intake — upstream of combustion, without modifying the engine.

Why hydrogen helps diesel combustion Diesel engines suffer incomplete combustion — fuel-rich zones exist where droplets haven't fully vaporised when combustion begins. Unburnt fuel exits as CO, HC, PM and wasted energy. Hydrogen ignites 12× more easily than diesel vapour. Its flame speed is 7× faster. Introduced in small quantities, it creates ignition kernels throughout the combustion chamber, igniting fuel-rich zones that diesel flame alone would miss. Result: more complete combustion from the same quantity of diesel. Less fuel consumed per kilometre. Cleaner exhaust. Cleaner engine.	The CHFA Zone B principle — critical for credibility Our published research identifies three operational zones: • Zone A: Too little hydrogen — combustion unchanged. • Zone B (optimal): Correct dosing — measurable improvement in every engine. • Zone C: Over-dosing — efficiency degrades. Advanced combustion phasing works against the engine. Our adaptive control system maintains Zone B dosing dynamically across load, RPM, engine age, and BS-norm. Newer BS-VI engines require narrower calibration than older BS-IV/III engines — our CHFA electronics account for this.
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System Specifications

Vehicle Types	Trucks (LCV to HCV), buses, tippers, dumpers, tankers, mining vehicles — any commercial diesel engine
Voltage Platforms	12V (LCVs, smaller vehicles) and 24V (HCVs, buses, construction equipment)
Installation	Retrofit to air intake only. No engine modification. No exhaust system changes. Typical install: 2–4 hours per vehicle.
Hydrogen Storage	Zero. On-demand generation only. No ADR/PESO compliance required. No explosion risk.
Input	Water + electricity from vehicle's own alternator. No chemicals, no urea, no consumables.
Connectivity	IoT-enabled. SGT GreenVision Digital Twin platform — real-time fleet monitoring, HHO performance tracking, driver behaviour correlation.
IP Status	Patent-applied. Vibration-resistant. Designed for Indian road conditions — heat, dust, potholes, 24/7 operation.

3. The Research Foundation — Published & NABL-Verified

SGT's research programme covers both controlled laboratory testing and real-world fleet validation. The CHFA research paper was co-developed with inputs from Automotive Research Association of India (ARAI) — India's apex automotive testing and standards body under the Government of India — and tested across all major diesel engine generations.

CHFA Research Paper — Key Findings Across All Diesel Engine Types

Tested on 5 platforms: Caterpillar C32 TA (1010 kVA), Cummins 380 kVA, Cummins 1500 kVA, 250 kVA genset, and 2023-model KOEL 500 kVA (BS-VI equivalent). Emission testing by NABL-accredited laboratories.

Parameter	Older Engines (CPCB I/II, BS-III/IV)	Newer Engines (CPCB IV / BS-VI)
Fuel Efficiency (kWh/L)	8–15% improvement at 40–60% load	4–13% improvement at 40–60% load
BSFC Improvement	10–10.6% reduction (CPCB II)	3.8–5.8% reduction (CPCB IV)
PM Reduction	Up to 80.1%	Positive across all loads at optimal dosing
NOx Reduction	Up to 34.6%	Up to 29.9%
CO Reduction	Up to 41.4%	Up to 15.1%
CO ₂ Reduction	Up to 42.9%	Up to 20.8%
SO ₂ Reduction	Up to 75.2%	Up to 63.6%
THC Reduction	Up to 48.9%	Up to 9.3%
Key Insight	Broader Zone B window — more tolerant of HHO dosing	Narrower Zone B — requires tighter adaptive control

Note: Research was conducted on diesel generator engines which share the same fundamental combustion architecture as vehicle diesel engines. Emission testing conducted by Ekdant Enviro Services (NABL), Lata Envirotech Services, and Neetal Laboratories (NABL, ilac-MRA) — all NABL-accredited to applicable IS standards.

The Vehicle Application — Why the Same Physics Applies

Commercial diesel vehicle engines (BS-III through BS-VI) operate on the same diesel combustion cycle as tested platforms. The Zone B enhancement mechanism — ignition acceleration by hydrogen, improved combustion completeness, exhaust O₂ increase — is engine-architecture independent. The adaptive CHFA control calibration for vehicles accounts for:

- Variable load profiles (unlike stationary DG at fixed load, trucks have highly variable load)
- Vibration, temperature fluctuation, and 24/7 duty cycles
- BS-VI OBD-2 monitoring — system is designed to be invisible to engine management systems
- Both 12V and 24V electrical architectures

4. Customer Evidence — Real Vehicles, Real Routes, Real Data

Every result presented in this section is from a customer-issued letter on company letterhead, based on actual diesel top-up records, CAN bus data, or operational mileage logs. SGT has not adjusted or reinterpreted these figures.

Enterprise POC: Coca-Cola Vendor Fleet (4-Month Controlled Trial)

4-month POC completed. 6–9% fuel reduction from baseline confirmed across vehicles.

The Coca-Cola POC was conducted under enterprise fleet management conditions with controlled baselines, standard routes, and monitored fuel consumption. The 6–9% improvement from baseline represents a conservative, real-world result with human drivers — not a laboratory outcome.

Current status: SGT is in active discussion with Coca-Cola for rollout across their vendor ecosystem — a potential multi-vehicle deployment.

Independent Customer Validation Letters

Customer	Vehicle	Baseline Mileage	Post-Install Result	Verified Saving
BLR Logistik (I) Ltd	Ashok Leyland NL01 AE0983 (CAN data, 5,382 km)	3.88 KMPL (14,920 km baseline)	4.75 KMPL (Phase 2, 23 days)	22% fuel improvement (CAN-verified)
BLR Logistik (I) Ltd	MH04 KU 3381 (Phase 1 trial)	3.19 KMPL (10,241 km baseline)	3.91 KMPL (5,792 km test run)	23% improvement (Phase 1)
Vasantara Logistics	Ashok Leyland GJ06 BV5314 (diesel top-up records)	3.5–3.9 KMPL (baseline monitored)	4.13–4.75 KMPL (22,580 km tested)	18–22% saving (Nov 2024–Apr 2025)
Narendra Roadlines	TATA SIGNA 4018.S BSVI GJ06 BT6413	3.1–3.3 KMPL (baseline)	3.7–3.9 KMPL (17,836 km tested)	14–17% saving (Nov 2024–Mar 2025)
Ranjit Petroleum	TATA SIGNA 4018.S BSVI GJ06 BT6108	3.4–3.6 KMPL (baseline)	4.0–4.35 KMPL (2,417 km tested)	18–21% saving (Feb–Apr 2025)
Coca-Cola Vendor Fleet	Multi-vehicle enterprise POC (controlled conditions)	Documented baseline (controlled protocol)	Controlled fleet measurement (4 months)	6–9% fuel reduction from baseline

All results are from customer-issued letters on company letterhead, dated April 2025. Vehicle registration numbers, baseline km, and diesel top-up logs are available for due diligence review.

An important note on result variability — and why we expect it

The customer results above range from 6% to 23%. This is not inconsistency — it is the honest outcome of human factors. BLR's 22–23% results were achieved on carefully monitored vehicles with disciplined fuel top-up records. Coca-Cola's 6–9% POC was conducted with a standard driver pool under real logistics conditions.

SGT's commercial model accepts this variability. At a product cost below 4% of annual fuel spend, even the lowest real-world saving (6%) delivers ROI well within 12 months. Better outcomes — 14–22% — are achieved with driver training and IoT monitoring through GreenVision.

5. International Market Presence

GreenDrive has been trialled and ordered across three international markets — demonstrating that the technology's applicability extends beyond Indian fleet conditions.

Market	Status	Context & Learning
Malaysia	Trial order completed	Validated product function in a Southeast Asian logistics context with different engine profiles and fuel grades. Operational learning incorporated into current product generation.
Uganda	Trial order completed	Demonstrated applicability in an African market context with older vehicle fleet profiles and variable fuel quality — conditions where Zone B benefit is broader.
Australia	Trial order completed	High-compliance market test. Validated product safety, installation protocol, and measurement methodology under stricter regulatory expectations.

SGT is deliberately cautious about international scaling in this category — payment systems, driver behaviour management, and service infrastructure vary significantly across markets. The strategy is to prove deeply in core markets before scaling internationally.

6. Global Research Evidence — HHO in Diesel Vehicle Engines

The performance of HHO-enriched diesel combustion in vehicle engines has been independently validated by research institutions across multiple countries. The following summarises the most relevant findings:

Metric	Finding	Source
Fuel Consumption	11.7–12.3% reduction on commercial vehicles (NEDC standardised road test)	<i>ScienceDirect — NEDC driving cycle study</i>
BSFC	12.6–13.7% average reduction; up to 20% at optimised flow rates	<i>Multiple controlled engine studies</i>
Engine Thermal Efficiency	Up to 31.5% improvement — more power per litre of diesel	<i>Peer-reviewed international studies</i>
PM / Smoke Opacity	25–95% reduction depending on HHO rate and load; 26.8% under NEDC cycle	<i>University of Technology Baghdad; multiple sources</i>
CO Emissions	13–54.5% reduction across studies	<i>ScienceDirect, MDPI, Springer</i>
HC Emissions	7.7–46.2% reduction	<i>Multiple peer-reviewed sources</i>
NOx Emissions	Reductions at moderate load at optimal dosing; increase possible at high load if over-dosed	<i>IJHE, ScienceDirect — consistent with Zone B/C model</i>
Long-Term Safety	500-hour continuous dual-combustion study — zero structural damage to engine components	<i>Peer-reviewed durability study</i>
Real-World Validity	40km extra-urban driving cycle road tests confirm lab results translate to operating conditions	<i>International road test validation studies</i>

Research institutions contributing to this evidence base include University of Technology Baghdad, UNAM Mexico, Escuela Superior Politécnica del Litoral Ecuador, and research groups in India, Turkey, Egypt, and Malaysia. Published in ScienceDirect, Springer, MDPI, SAGE, Elsevier, and Nature Scientific Reports.

7. The ROI Framework — Designed for Certainty

SGT's commercial model for logistics is built on one non-negotiable principle: the system must cost less than 4% of the vehicle's annual fuel expenditure — ensuring every operator recovers the investment within 12 months even at the lowest observed savings.

Annual Fuel Cost Framework (India)

Based on diesel at ₹90/litre and a 300 working-day year:

Vehicle Profile	Annual Fuel Spend	4% Threshold (Max System Cost)	Saving @ 6% (Conservative Real-World)
Long-haul HCV (1500 L/month)	₹16.2 L	₹65,000	₹97,200
Heavy truck (2500 L/month)	₹27 L	₹1.08 L	₹1.62 L
Fleet vehicle (4000 L/month)	₹43.2 L	₹1.73 L	₹2.59 L
Mining/construction (8000 L/month)	₹86.4 L	₹3.46 L	₹5.18 L
Fleet of 50 trucks (2500 L/truck/month)	₹13.5 Cr	₹54 L	₹81 L

L = Lakhs, Cr = Crores. 'Max System Cost' = maximum product price at 4% threshold. 'Saving @ 6%' = annual saving if only 6% fuel reduction achieved. Actual savings reported by customers range 6–22%.

The 1,500 litre/month minimum — why SGT sets this threshold

Below 1,500 litres per month (~50 litres/day), the absolute fuel saving is too small to produce a measurable, unambiguous ROI signal above the noise of normal fuel variation, driver behaviour, and route differences. SGT only deploys GreenDrive on vehicles consuming ≥1,500 litres/month — this ensures every customer can clearly verify their return.

The Additional Benefits That Compound Over Time

Fuel saving is the primary measured ROI metric. But GreenDrive delivers three further benefits that generate ongoing value:

 Engine Decarbonisation	Continuous HHO-assisted combustion burns off carbon deposits on pistons, valves, and injector tips during normal operation — no downtime, no manual cleaning. Cleaner engine = better fuel atomisation = better efficiency over time. Extended TBO (time between overhauls) is a real, compounding maintenance saving.
 Cleaner Exhaust	CO, PM, HC, and NOx reductions mean vehicles pass pollution checks more easily and generate less health-damaging tailpipe emissions. In BS-VI non-compliant retrofitted trucks, this is an additional compliance protection.
 GreenVision IoT Platform	Real-time vehicle monitoring identifies fuel theft patterns, driver behaviour deviations, and HHO system performance. Fleets using GreenVision consistently report higher savings because human factors are managed — not ignored.

8. The Investor Perspective — Why This Category Deserves Re-Evaluation

The conventional investor hesitation about logistics HHO is rational — but based on the previous generation of the technology.

Investors have stayed away from logistics HHO for three reasons: unverifiable claims, driver interference, and no independent validation. SGT has systematically addressed each of these:

Investor Concern	SGT's Response
"Claims are unverifiable"	We have customer letters on letterhead from BLR Logistik (30+ year logistics company), Vasantara Logistics, Narendra Roadlines, and Ranjit Petroleum — all with specific vehicle numbers, baseline KMs, and diesel top-up records. A completed enterprise POC with Coca-Cola provides the highest-credibility reference point available.
"Driver behaviour destroys savings"	We agree — which is why our commercial model guarantees ROI below 4% of annual fuel spend even if the driver negates part of the saving. GreenVision IoT monitors driver behaviour and fuel theft, allowing fleet managers to intervene. We don't pretend human factors don't exist.
"No independent scientific validation"	The CHFA research paper, developed with ARAI input and tested across 5 engine platforms by NABL-accredited laboratories, provides institutional scientific validation independent of customer-reported results.
"Outcome ambiguity — actual vs claimed"	SGT does not claim 20% savings. We claim 6–9% as our enterprise POC baseline, and price the product to deliver ROI at even lower savings. Higher outcomes are customer-documented bonuses, not the value proposition.
"Fragmented market, poor unit economics"	We focus on vehicles consuming $\geq 1,500$ L/month — commercial fleets, not individual truck owners. This narrows our target to operators with measurable baselines, payment discipline, and fleet management infrastructure.
"No sustainable competitive moat"	Patent-applied CHFA technology. Proprietary Zone B adaptive control. Published research. ARAI engagement. 3+ year head start. The only non-Chinese industrial-grade HHO manufacturer with multi-category proven technology.

The Market Opportunity

- India has **3.5 million+** registered commercial vehicles consuming $>1,500$ L/month of diesel.
- Fuel represents **45–60%** of total fleet operating cost — the single largest controllable expense.
- EV transition for HCV is **10–15 years away** in India at commercial scale — the existing diesel fleet needs bridging solutions now.
- The Coca-Cola POC positions SGT for **vendor fleet aggregation** — one enterprise client = access to hundreds of vendor vehicles.

9. Emissions & ESG — The Coupled Benefit

Every litre of diesel saved eliminates 2.68 kg of CO₂ from the atmosphere. Fuel savings and Scope 1 emissions reduction are mathematically identical. GreenDrive does not merely reduce regulated pollutants — it reduces the amount of fossil fuel consumed per kilometre driven.

Emission	Reduction Range	Mechanism / Reference
CO₂ (Scope 1)	Proportional to fuel saved (6–22% observed)	Direct: less diesel burned = proportionally less CO ₂ . No post-treatment — reduction is at combustion source.
Particulate Matter (PM2.5)	Up to 80.1% (NABL-tested)	Improved combustion completeness — fewer carbon particles formed. Most visible immediate benefit.
Carbon Monoxide (CO)	Up to 41.4% (NABL-tested)	More complete oxidation of CO to CO ₂ . Direct product of improved combustion completeness.
NOx	Up to 34.6% (at optimal dosing)	Lower combustion temperature peaks at Zone B dosing. Increase risk if over-dosed (Zone C) — adaptive control prevents this.
SO₂	Up to 75.2% (NABL-tested)	Indirect: improved combustion completeness reduces sulphur compounds in exhaust.
Total Hydrocarbons	Up to 48.9% (NABL-tested)	Unburnt hydrocarbons dramatically reduced as combustion completeness improves.

Note: All emission reduction figures are from NABL-accredited laboratory testing of diesel generator engines using the same combustion architecture as vehicle diesel engines. Vehicle-specific emission testing is ongoing.

BRSR / ESG Reporting Value for Fleet Operators

Listed companies with logistics operations are now required to report Scope 1 emissions under SEBI's BRSR (Business Responsibility and Sustainability Report) mandate. GreenDrive provides:

- Documented, measurable, per-vehicle Scope 1 CO₂ reduction — directly reportable
- IoT data via GreenVision for audit-grade emissions tracking per vehicle, per route, per month
- No capital expenditure on fleet replacement — immediate deployment on existing vehicles
- Contribution to Scope 3 emission reduction for enterprise clients (Coca-Cola, FMCG, manufacturing) managing vendor fleet emissions

10. Implementation — How We Deploy GreenDrive

SGT's deployment methodology is designed to manage the human variables that define success in logistics applications.

#	Stage	Activity
1	Baseline Assessment	Minimum 4-week fuel consumption baseline using diesel top-up records or CAN data. Route mapping, load profile, and driver behaviour baseline established.
2	System Installation	GreenDrive fitted to vehicle air intake. Full electrical integration with 12V or 24V system. GreenVision IoT module connected. Installation: 2–4 hours/vehicle. No downtime from regular operations.
3	Calibration & Zone B Optimisation	HHO flow rate calibrated to engine type, age, BS-norm, and primary load profile. Zone B confirmed through exhaust O ₂ /CO ₂ monitoring. Adaptive control parameters set.
4	Monitoring Period (4–8 Weeks)	GreenVision tracks fuel consumption, HHO performance, and driver behaviour in real time. Baseline comparison generated weekly. Fuel saving verification built into the monitoring dashboard.
5	Validation Report	Customer receives documented fuel saving report — vehicle-specific, period-specific, verified against diesel top-up records. KMPL improvement and CO ₂ reduction quantified.
6	Fleet Scaling	Based on validated results, deployment extended to full fleet. GreenVision fleet-level reporting for BRSR/ESG compliance.

Target Vehicle Profile

- **Fuel consumption:** ≥1,500 litres/month (≥50 litres/day)
- **Engine types:** BS-III, BS-IV, BS-VI diesel — trucks, tippers, tankers, buses, construction, mining
- **Fleet structures:** Minimum 5-vehicle fleets preferred for GreenVision baseline integrity
- **Enterprise clients preferred:** Fleet operators, 3PLs, FMCG/pharma vendor fleets, mining/construction contractors, state transport

11. In Summary — The SGT GreenDrive Position

What we guarantee	What our customers actually achieve
• ROI within 12 months (product cost < 4% of annual fuel spend) • Measurable fuel saving on every engine operating in Zone B • CO₂ reduction proportional to fuel saved — directly reportable Scope 1 • Cleaner engine through continuous decarbonisation • Emission reductions validated by NABL-accredited science	• BLR Logistiks: 22–23% fuel improvement (CAN-verified) • Vasantara Logistics: 18–22% saving (22,580 km tested) • Narendra Roadlines: 14–17% saving (17,836 km tested) • Ranjit Petroleum: 18–21% saving (diesel top-up verified) • Coca-Cola Enterprise POC: 6–9% reduction from controlled baseline

Ready to validate GreenDrive on your fleet?

We offer a structured 4-week baseline assessment + installation + monitored validation protocol. You see the numbers on your vehicles, on your routes, with your drivers — before committing to full fleet deployment.

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